

Array size deduction in new-expressions

Timur Doumler (papers@timur.audio)

Document #: P1009R1
Date: 2018-11-26
Project: Programming Language C++
Audience: Core Working Group

Abstract

In this paper we propose to fix a particular inconsistency in the initialization rules of C++ by allowing array size deduction in *new-expressions*. This aligns their behaviour with that of initialization everywhere else in the language.

1 Motivation

Bjarne Stroustrup pointed out the following inconsistency in the C++ language:

```
double a[]{1,2,3};           // this declaration is OK, ...  
double* p = new double[]{1,2,3}; // ...but this one is ill-formed!
```

Jens Maurer provided the explanation why it doesn't work: For a *new-expression*, the expression inside the square brackets is currently mandatory according to the C++ grammar. When uniform initialization was introduced for C++11, the rule about deducing the size of the array from the number of initializers was never extended to the *new-expression* case. Presumably this was simply overlooked. There is no fundamental reason why we cannot make this work.

Admittedly, deducing the array size in a *new-expression* is code that probably only very few people would actually write. One could therefore argue that this is a problem not worth fixing.

However, when teaching C++ initialization rules, we observe that most people intuitively expect uniform initialization in a *new-expression* to follow the same rules as uniform initialization everywhere else in the language. This exception is very unfortunate and tends to upset and surprise people when pointed out to them.

The existence of such exceptions is exactly the reason why C++ initialization rules are so notorious for being complicated, and why most C++ developers struggle with them. There are just too many non-obvious inconsistencies. We therefore propose to remove this particular one—not because this is a problem that people would frequently run into (they don't), but because fixing it is straightforward, the fix is a pure extension that does not impact any other part of the standard, and it would make initialization rules in C++ simpler, more uniform, and easier to teach.

2 Solution

We propose to allow omitting the array bound in a *new-expression*, as long as the array size can be deduced from the initializer list, in the same way it is already allowed in regular array declarations:

C++17	This proposal
<code>double a[]{1,2,3};</code> // OK	<code>double a[]{1,2,3};</code> // OK
<code>double* p = new double[]{1,2,3};</code> // Not OK	<code>double* p = new double[]{1,2,3};</code> // OK

A special case are arrays with no elements. While a declaration of an object of such type is ill-formed, it is fine to allocate one in a *new-expression*:

```
int a[0]{};           // this declaration is ill-formed, ...
int* p = new int[0]{}; // ...but this one is OK!
```

This keeps consistency with C, where `malloc(0)` returns a (non-dereferenceable) pointer, and is occasionally useful in C++, e.g. in templates where the array size is a non-type template parameter. To be maximally consistent, we propose that an array size of 0 in a *new-expression* should be deduced if the initializer consists of empty braces:

C++17	This proposal
<code>double* p = new double[0]{};</code> // OK	<code>double* p = new double[0]{};</code> // OK
<code>double* p = new double[]{};</code> // Not OK	<code>double* p = new double[]{};</code> // OK

Here, both versions (with or without the 0) would have the same effect. This way, array size deduction in *new-expressions* behaves the exact same way for any array size that is allowed in a *new-expression*.

3 Proposed wording

The reported issue is intended as a defect report with the proposed resolution as follows. The effect of the wording changes should be applied in implementations of all previous versions of C++ where they apply. The changes are relative to the C++ working paper [Smith2018].

Modify [expr.new] paragraph 1 as follows:

```
noptr-new-declarator :
    [ expressionopt ] attribute-specifier-seqopt
    noptr-new-declarator [ constant-expression ] attribute-specifier-seqopt
```

Modify [expr.new] paragraph 6 as follows:

Every *constant-expression* in a *noptr-new-declarator* shall be a converted constant expression of type `std::size_t` and shall evaluate to a strictly positive value. ¶If the *expression* in a *noptr-new-declarator* is present, it is implicitly converted to `std::size_t`. [Example: Given the definition `int n = 42`, `new float[n][5]` is well-formed (because `n` is the *expression* of a *noptr-new-declarator*), but `new float[5][n]` is ill-formed (because `n` is not a constant expression). — end example]

If the *type-id* or *new-type-id* denotes an array type of unknown bound, or the *noptr-new-declarator* syntax is used and the *expression* in a *noptr-new-declarator* is omitted, a *new-initializer* shall be provided and shall be a brace-enclosed *initializer-list* containing `n` *initializer-clauses*. In this case the allocated object is an array with `n` elements, and if `n` is greater than zero, the elements of the array are initialized by the *initializer-list* as described in [dcl.init.aggr].

4 Previous work and committee guidance

R0 of the present paper was initially targeting C++20. It was reviewed by both EWGI and EWG in San Diego. The design was modified to include the special case of arrays with no elements. EWG then voted to make the proposal a defect report against C++ and to forward it to CWG. The present R1 reflects all of these changes.

After R0 was published, it was pointed out that Ville Voutilainen already had written an earlier paper about this issue [P0965]. However to our knowledge, that paper was never discussed in WG21.

Ville’s paper mentions the inconsistency we discuss here, however does not propose a technical solution for it (which we do here). It also mentions a second inconsistency, but then admits that “may be beyond fixing” and would require modifying the grammar in what “is certainly a breaking change”. That second inconsistency is a much more exotic corner case and we do not discuss it here.

Document history

- **R0**, 2018-10-08: Initial version
- **R1**, 2018-11-26: Added discussion of no-elements case; revised wording; made proposal a DR; added reference to P0965.

Acknowledgements

Many thanks to Richard Smith and Tim Song for their help with the wording, and to JF Bastien and Patrice Roy for their comments.

References

- [P0965] Ville Voutilainen. Initializers of objects with automatic and dynamic storage duration have funny inconsistencies. <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0965r0.html>, 2018-02-12.
- [Smith2018] Richard Smith. Working Draft, Standard for Programming Language C++. <https://github.com/cplusplus/draft>, 2018-10-08.